Quantification: 536-63x_Hyp_RD_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	21
interior lacuna count	21
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 21)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2913.24	2738	1640.9	px^2
major_axis_length_px	lacuna shape	100.689	102.242	39.0576	px
minor_axis_length_px	lacuna shape	42.0044	39.5227	14.1203	px
aspect_ratio	lacuna shape	2.4461	2.126	0.8775	unitless
eccentricity	lacuna shape	0.8764	0.8825	0.0764	unitless
solidity	lacuna shape	0.7734	0.8154	0.1295	unitless
roots_count	headline	10.619	10	5.536	count
ring_length_r30_px	headline	315.905	325	161.083	px
ring_length_r60_px	reported	643.048	601	323	px
owned_length_px	ownership-dependent	821.57	842.269	592.21	px
edge_count	ownership-dependent	40.7143	38	28.9277	count
mean_edge_length_px	ownership-dependent	20.638	19.753	6.284	px
perimeter_px	normalised	326.693	298.534	145.922	px
ring_area_r30_px2	normalised	9632.76	10082	3167.61	px^2
ring_area_r60_px2	normalised	21496.1	22623	7046.38	px^2
in_frame_fraction_r30	normalised	0.9877	1	0.0427	unitless
in_frame_fraction_r60	normalised	0.9754	1	0.0764	unitless
ring_density_r30	normalised	0.0314988	0.031131	0.00828807	px^-1
ring_density_r60	normalised	0.028746	0.0286761	0.00751178	px^-1
roots_per_100px_perimeter	normalised	3.301	3.3497	1.0422	per 100 px
ring_attached_length_r30_px	network v2	282.333	273	156.25	px
ring_attached_length_r60_px	network v2	496.619	470	286.687	px
ring_length_w_r30_px	network v2	356.762	368.291	180.447	px
ring_length_w_r60_px	network v2	731.604	682.541	362.432	px
sholl_crossings_r10	network v2	7.9524	8	3.3982	count
sholl_crossings_r20	network v2	8.5238	9	4.0202	count
sholl_crossings_r30	network v2	8.4286	8	3.9443	count
loop_count	loops, report only	2.0952	1	2.7732	count
loop_edge_share	loops, report only	0.1354	0.1316	0.1131	share
ring_width_mean_r30_px	width	7.5308	7.3668	0.5841	px

Per field
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measure	value	units
analysed_area_px2	987398	px^2
total_skeleton_length_px	25420	px
canalicular_length_density_per_px	0.0257444	px^-1
junction_count	497	count
junction_density_per_px2	0.00050334	px^-2
lacuna_count	21	count

lacunae_per_px2	2.127e-05	px^-2
skeleton_component_count	364	count
mean_component_length_px	69.8352	px
median_component_length_px	23.5	px
field_length_density_w_per_px	0.0292375	px^-1
field_density_without_flagged_per_px	0.0269446	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	247	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0257444	px^-1
field_length_density_with_unattached_per_px	0.0259946	px^-1
loop_count	55	count
loop_edge_share	0.1542	share
edge_count	1342	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8293	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25217	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2181	148.407	31.7971	4.6673	0.9768	0.5491	-0.2241	127.792	674.16	False
2	3003	146.448	39.2919	3.7272	0.9633	0.5868	-0.1342	219.3	634.066	False
3	4348	102.17	64.0204	1.5959	0.7793	0.7299	-0.8016	123.509	117.145	False
4	6103	130.52	74.6261	1.749	0.8204	0.7441	-0.6975	211.602	115.526	False
5	4713	113.091	66.6058	1.6979	0.8082	0.6869	0.0149	166.949	811.934	False
6	2838	120.116	56.4992	2.126	0.8825	0.5006	-0.0922	183.402	482.588	False
7	593	40.0782	21.7723	1.8408	0.8396	0.7583	0.4372	249.45	834.877	False
8	5444	169.703	43.0843	3.9389	0.9672	0.8492	-0.1663	315.438	989.012	False
9	5032	145.049	50.087	2.8959	0.9385	0.6975	-0.3567	389.803	259.63	False
10	1325	56.519	31.2238	1.8101	0.8335	0.9138	-0.4181	431.483	840.339	False
11	1869	63.382	43.4291	1.4594	0.7284	0.783	-0.4231	489.012	823.251	False
12	4023	130.997	44.5634	2.9396	0.9404	0.8529	-0.2947	478.273	373.626	False
13	2063	91.1753	31.0575	2.9357	0.9402	0.8892	-0.3885	511.078	505.802	False
14	2736	102.242	36.178	2.8261	0.9353	0.8653	-0.4465	632.235	931.07	False
15	2738	96.5558	37.4594	2.5776	0.9217	0.8784	-0.3793	679.917	399.446	False
16	3288	105.667	42.059	2.5123	0.9174	0.8761	-0.3815	724.683	177.336	False
17	797	46.6745	23.1748	2.014	0.868	0.8935	-0.2375	760.548	828.605	False
18	584	33.791	23.1054	1.4625	0.7297	0.9125	-0.7852	793.788	813.11	False
19	1183	68.5981	39.5227	1.7357	0.8173	0.5796	-0.4587	845.742	872.086	False
20	4334	128.945	44.2583	2.9135	0.9393	0.8796	-0.1396	903.815	374.562	False
21	1983	74.3475	38.2775	1.9423	0.8573	0.8154	-0.0273	970.974	775.836	False

Network per lacuna

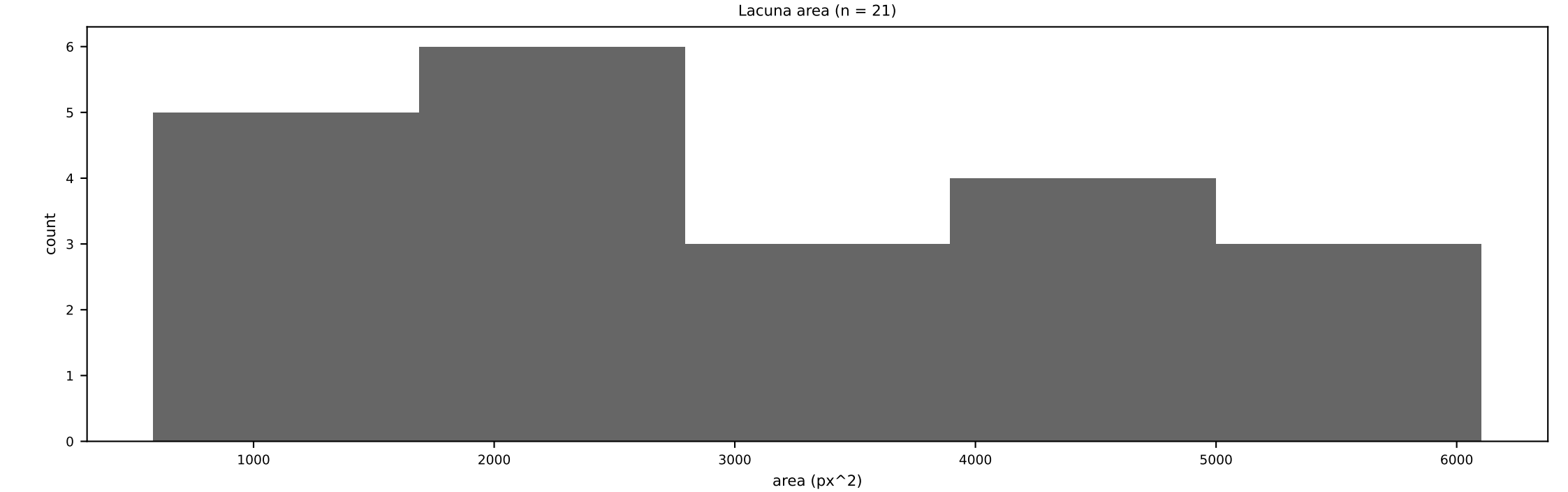
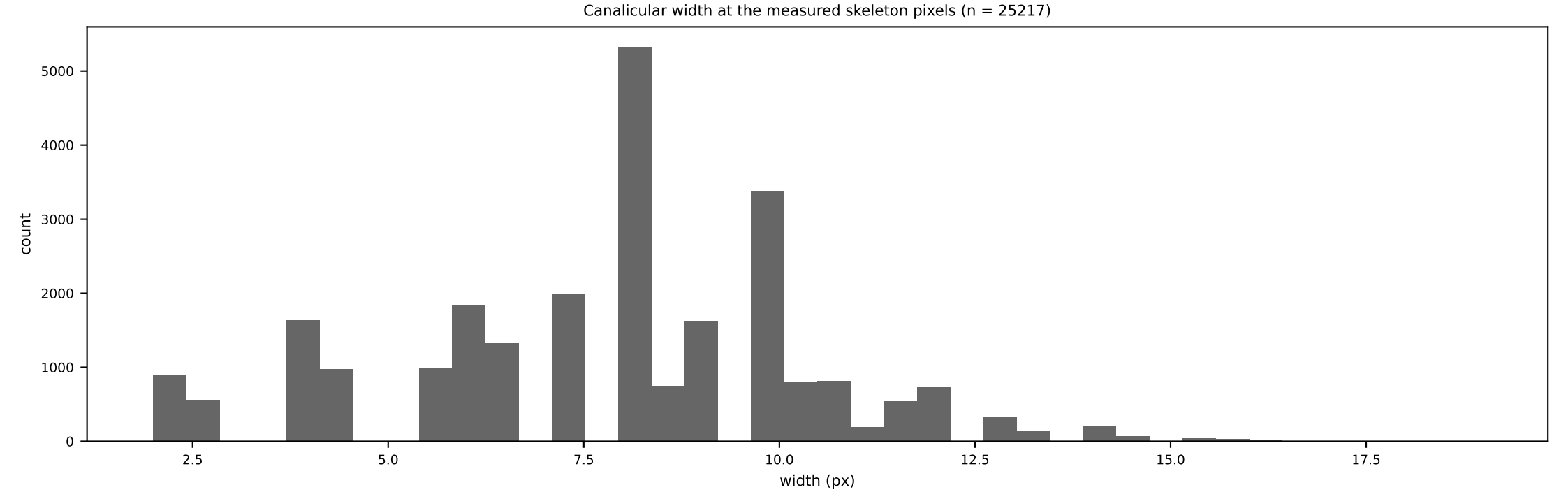
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	21	392	645	542.245	38	14.2696	7.9442
2	14	325	728	943.122	56	16.8415	7.2425
3	8	341	587	577.387	31	18.6254	7.2773
4	10	364	847	1131.56	51	22.1875	7.3668
5	20	457	794	1082.08	45	24.0461	7.528
6	19	723	1433	1324.53	81	16.3522	6.6187
7	6	202	455	726.612	31	23.4391	7.3169
8	10	325	580	1119.45	52	21.5278	7.3803
9	20	632	1263	2681.43	133	20.1611	7.0906
10	4	90	197	79.1838	7	11.312	7.454
11	8	165	372	227.338	12	18.9448	6.9494
12	13	411	890	1092.82	61	17.9151	8.8041
13	10	286	601	967.637	42	23.039	8.2735
14	6	200	473	298.563	17	17.5626	7.9746
15	11	305	670	1297.54	48	27.0322	8.1069
16	10	349	860	1189.62	50	23.7924	7.3406
17	4	91	217	232.208	16	14.513	6.8986
18	2	96	145	82.9117	2	41.4558	7.9853
19	11	289	439	498.416	33	15.1035	7.1548
20	10	375	855	842.269	33	25.5233	8.5809
21	6	216	453	316.049	16	19.753	6.858

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	415.641	10082	20733	1	1	0.0388812	0.0311098	5.0524
2	495.973	10286	21637	1	1	0.0315963	0.0336461	2.8227
3	488.535	9556	20470	1	1	0.0356844	0.0286761	1.6375
4	609.269	12518	25962	1	0.9991	0.0290781	0.0326246	1.6413
5	442.446	12800	26586	1	1	0.0357031	0.0298653	4.5203
6	514.931	14099	31464	1	1	0.0512802	0.0455441	3.6898
7	128.539	4684	10910	1	1	0.0431255	0.0417049	4.6678
8	423.747	11743	23622	0.8152	0.7003	0.0276761	0.0245534	2.3599
9	487.973	14467	32198	1	1	0.0436856	0.039226	4.0986
10	156.16	5263	12719	1	1	0.0171005	0.0154886	2.5615
11	228.202	6707	15935	1	1	0.0246012	0.0233448	3.5057
12	348.433	12410	27080	1	1	0.0331185	0.0328656	3.731
13	227.38	9187	22623	1	1	0.031131	0.0265659	4.3979
14	274.735	10452	25593	1	1	0.0191351	0.0184816	2.1839
15	250.693	9821	24082	1	1	0.0310559	0.0278216	4.3878
16	298.534	10681	26849	1	1	0.0326748	0.032031	3.3497
17	131.776	4471	9578	1	1	0.0203534	0.0226561	3.0355
18	102.847	3623	7408	1	1	0.0264974	0.0195734	1.9446
19	259.557	9006	18739	1	1	0.0320897	0.0234271	4.238
20	340.877	12243	29334	1	0.982	0.0306298	0.0291471	2.9336
21	234.309	8189	17896	0.9258	0.8017	0.0263769	0.0253129	2.5607

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	342	438	441.747	736.24	8	12	10
2	313	588	369.563	829.24	8	8	7
3	273	470	389.676	672.541	10	11	9
4	350	640	410.676	970.105	13	11	8
5	428	723	517.546	907.98	10	10	12

6	694	1052	813.127	1613.81	14	11	16
7	202	435	228.924	526.002	6	3	5
8	325	552	368.291	663.198	9	10	7
9	563	1136	709.985	1430.87	13	16	16
10	76	76	102.326	221.995	4	0	2
11	150	205	184.196	422.818	3	5	5
12	371	812	458.635	1009.29	8	11	13
13	243	396	326.735	682.541	12	7	10
14	149	242	228.865	544.571	6	9	7
15	305	602	350.978	779.181	8	9	8
16	261	536	373.551	951.796	8	11	11
17	91	197	104.083	252.622	3	3	2
18	47	56	109.983	170.409	3	4	5
19	272	404	334.635	504.073	7	7	7
20	329	642	423.019	955.595	10	15	12
21	145	227	245.451	518.801	4	6	5

536-63x_Hyp_RD_z08c1-2: distributions, pre-validation, px